

*BRAC*'s

**Vishwakarma Institute of Technology, Pune**

**Practical Implementation Sheet**

Department: CSE-AI      Semester: 5      Academic Year: 2026-27

Division: E      Batch: 2      Practical No: 4

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**Subject Name : Deep Learning**

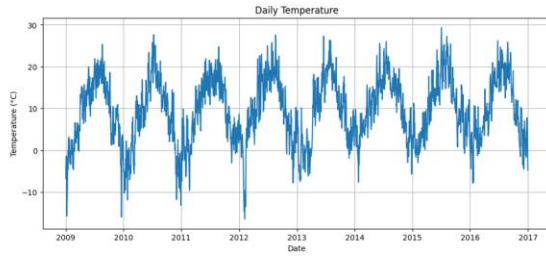
**Problem Statement**

Develop an LSTM-based model for time-series forecasting using a weather dataset. Use historical weather observations to predict the next day's temperature.

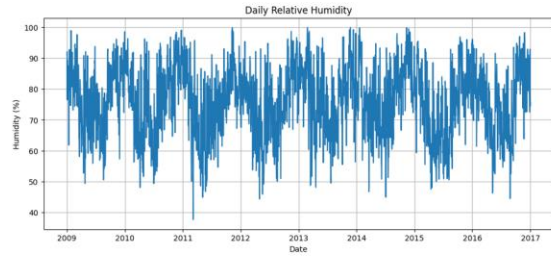
**Algorithm**

1. Import the required libraries such as NumPy, Pandas, Matplotlib, TensorFlow/Keras, and Scikit-learn.
2. Load the Jena Climate weather dataset containing timestamped weather observations.
3. Convert the Date Time column into a datetime format and resample the 10-minute observations into daily averages.
4. Select temperature, pressure, relative humidity, and wind speed as input features, with temperature as the forecasting target.
5. Split the daily data chronologically into 80% training data and 20% testing data.
6. Normalize the input features using MinMaxScaler fitted only on the training data.
7. Create 30-day sliding-window sequences, where the previous 30 days are used to predict the next day's temperature.
8. Construct an LSTM network with two LSTM layers, Dropout layers, and a Dense output layer.
9. Compile the model using the Adam optimizer and mean squared error loss.
10. Train the LSTM model for 20 epochs with a batch size of 32 and 10% validation split.
11. Generate temperature predictions for the test sequences and inverse-transform the predicted and actual values to degrees Celsius.
12. Evaluate the forecasting performance using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).
13. Plot the actual and predicted temperatures to visualize the forecasting performance.

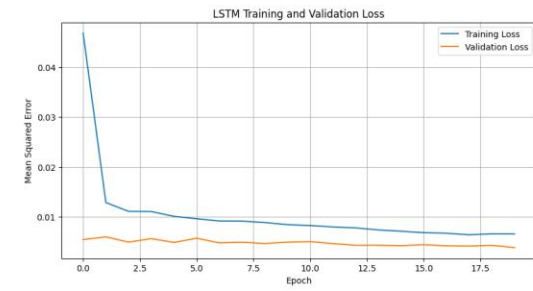
Daily Temperature



Daily Relative Humidity



Training and Validation Loss



Actual vs Predicted Temperature using LSTM

